

Formalizing Resistance

Aaron Avram

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1 Set up

The basic setup for the system is as follows:

- There is a group of agents G .
- A set of states S where $s \in S$ has probability $p(s)$ of occurring.
- For each $s \in S$ there is a set $A(s)$ of possible actions in state s .
- A policy is function that maps each $s \in S$ to a probability distribution over $A(s)$. Fix a dominant policy π_d .
- Each agent $i \in G$ has a fixed reward $u_i(s, a)$ for action $a \in A(s)$ for $s \in S$.
- Each agent i has an expected reward under a policy π in state s :

$$U_i(s, \pi) = \mathbb{E}_{a \sim \pi(s)} [u_i(s, a)]$$

- For each $i \in G$ there is a set of states $S_r^i \subseteq S$ where $U_i(s, \pi_d) \ll 0$.

We will assume the dominant norm stays fixed throughout all of the calculations for simplicity.

2 Defining Resistance Theoretically

We give a theoretical definition of a resistant subgroup G_r of G .

$$w_i(s) = \max(0, -U_i(s, \pi_d))$$

hence $w_i \in \mathbb{R}_{\geq 0}^{|S|}$. The probability function on S defines a norm on $\mathbb{R}^{|S|}$ given by:

$$\|\cdot\|_p : x \rightarrow \sqrt{\sum_{s \in S} p(s) x(s)^2}$$

For a subgroup $H \subseteq G$ define

$$D(H) = \sum_{i \in H} \|w_i\|_p^2$$

which is the total disagreement with π_d within H . If $\bar{w}_H$ is the mean w_i in H , we define a related quantity:

$$F(H) = \sum_{i \in H} \|w_i - \bar{w}_H\|_p^2$$

and by expanding the square this is equivalent to

$$F(H) = \frac{1}{2|H|} \sum_{i,j \in H} \|w_i - w_j\|_p^2$$

this is called the friction of H and represents how much cohesion the group H has in its disagreement with π_d .

Given a tolerance $\delta > 0$ we define the maximally resistant subgroup of G as

$$G_r = \operatorname{argmax}_{H \in \mathcal{P}(G)} D(H)$$

subject to the constraint

$$F(H)/|H| \leq \delta$$

where the constraint is necessary as $D(H)$ is monotone in H with respect to $\subseteq$.

If $\delta = 0$ then this requires that the maximally resistant subgroup is frictionless, i.e. its members perfectly align on their disagreement with π_d .

Now we define a similarity matrix for the set of agents.

$$K_{ij} = \exp(-\|w_i - w_j\|^2/2\sigma^2)$$

and note that Jensen's inequality gives the relation:

$$\mathbb{E}_H[K_{ij}] \leq \exp \mathbb{E}_H[\|w_i - w_j\|_p^2/2\sigma^2] = \exp\left(\frac{F(H)}{|H|\sigma^2}\right)$$

which gives the bound

$$|H|\sigma^2 \log \mathbb{E}_H[K_{ij}] \leq F(H)$$

To prove a flipped version of this inequality we assume that $|u_i(s, a)|$ is bounded by a uniform constant. This implies that there exists R such that for all $i, j \in H$

$$\|w_i - w_j\|_p^2 \leq R$$

let $M = \frac{R^2}{2\sigma^2}$. Then the function

$$1 - \exp(-x)$$

is concave so for $x \in [0, M]$

$$1 - \exp(-x) \geq x \left(1 - \exp(-M)\right)/M$$

rearranging we see that

$$x \leq M \frac{1 - \exp(-x)}{1 - \exp(-M)}$$

now set $x = \|w_i - w_j\|_p^2/2\sigma^2$ then

$$\|w_i - w_j\|_p^2/2\sigma^2 \leq \frac{R^2}{2\sigma^2} \frac{1 - K_{ij}}{1 - \exp(-M)}$$

or

$$\|w_i - w_j\|_p^2 \leq R^2 \frac{1 - K_{ij}}{1 - \exp(-M)}$$

hence

$$F(H) \leq \frac{R^2}{2} \frac{\frac{1}{|H|} - \mathbb{E}[K_{ij}]}{1 - \exp(-\frac{R^2}{2\sigma^2})}$$

in other words this implies that the expected pairwise distance is small whenever $F(H)$ is.

Since $F(H)$ would be hard to compute in practice K_{ij} can be used as an approximation.

3 Estimating Rewards

Each agent keeps a running average of their personal benefit under π_d (this defined similarly to the learning norms paper). Now I propose that each agent shares their estimates with all other agents. This forms a matrix:

$$V_i[j, s]$$

which tracks i 's estimate of j 's personal benefit under the dominant norm in state s . $V_i[j, s]$ should approach $U_j(s, \pi_d)$.

Currently I assume that the message passed from agent j to i is perfectly accurate. However, this could be extended in two ways: one, simulating imperfect communication by adding Gaussian noise to each message and two, simulating deception by allowing j to learn the message to pass to i that is most favourable to it.

Next we define

$$\hat{w}_j^i(s) = \max(0, -V_i[j, s])$$

under the above assumptions this is independent of i so we just write $\hat{w}_j(s)$. Additionally define

$$\hat{d}_{jk}^2 = \|\hat{w}_j(s) - \hat{w}_k(s)\|_p^2$$

and

$$\hat{K}_{jk} = \exp(-\hat{d}_{jk}^2/\sigma^2)$$

then

$$\epsilon_j = \|w_j - \hat{w}_j\|_p$$

with rewards bounded in absolute value by B Hoeffding's inequality gives the bound

$$|w_j(s) - \hat{w}_j(s)| < \sqrt{2 \log(2/\delta')/n_j(s)}$$

with probability $1 - \delta'$ where $n_j(s)$ is the number of times j has been in state s so far in the system. This then gives the bound

$$\epsilon_j < \sqrt{2|S| \log(2|S||N|/\delta')/n_j}$$

where n_j is the number of times agent j has acted so far.

This shows that ϵ_j can be bounded by quantities observable by the agents, but for simplicity I will assume below that each agent j has access to ϵ_j .

4 Formation

We now define conditions for resistant subgroup formation.

An agent i is called a candidate if

$$\|w_i\|_p^2 \geq \tau$$

for some fixed constant τ .

Then i admits j to its group if

$$\hat{d}_{ij} - \epsilon_i - \epsilon_j < r_{in}$$

and i leaves if some member j has

$$\hat{d}_{ij} - \epsilon_i - \epsilon_j > r_{out}$$

where $r_{out} > r_{in}$ for hysteresis.

These kinds of decisions would occur at update epochs like in the Learning Norms paper.